Vishnu N. Athreya

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EDUCATION

Michigan State University | B.S. Electrical Engineering **Expected May 2028**

GPA: 3.7 | Robotics & Smart Devices concentration | Business minor

Coursework: Digital Logic, Microcontrollers & Digital Systems, Electronic Circuits, Circuits & Systems I & II, Electromagnetism Waves & Fields I, EE Analysis, Circuits Lab, Electronic Circuits Lab

TECHNICAL SKILLS

Programming: Verilog, Python, C, C++, Java, MATLAB

Hardware & Design: Waveform debug, Vivado, KiCad, Altium, LTspice, PCB layout, FPGA Implementation

Tools: Docker, VS Code, Jira, Confluence, Copilot, Codex, Claude Code

ENGINEERING EXPERIENCE

Stoneridge Electronics **Novi, MI | May 2026 - Present**

Functional Safety Engineer (Part-Time); previously Functional Safety Engineering Intern

- Translated system, functional, and safety requirements into Hardware Requirements Specifications (HwRS), improving requirement quality, consistency, and traceability.
- Traced requirements, interfaces, verification, and safety artifacts to assess downstream impacts of customer-driven product, software, and UI changes.
- Modeled system boundaries and dependencies to trace hazards to requirements; standardized reusable templates across engineering teams.
- Developed ISO 26262 Item Definitions, HARA, and Safety Goals for MirrorEye to inform safety requirements and the Functional Safety Concept.
- Led a Copilot compliance assistant from development to deployment, integrating standards and enterprise knowledge to cut comparison time 83% and manual searches 75%.

TriMas Packaging | Process/Automation Intern **New Albany, OH | Jun 2025 - Aug 2025**

- Cut net rejects by 50% by diagnosing false suction rejects, standardizing assembly-to-test pushdown distance, and updating HMI/PLC logic using station data and part-quality tests.
- Analyzed 2 months of OEE data across 3 shifts to flag reject anomalies; supported troubleshooting and new product trials through root cause analysis and machine adjustments.
- Reduced training documentation by 80% across 4 production lines and 12 machines by applying Lean/Kaizen to rebuild SOPs and onboarding guides with process engineers.

ENGINEERING PROJECTS

8-bit ALU Design and Verification **ECE 331 | Oct 2026**

- Designed and implemented an 8-bit Verilog ALU on an FPGA using eight bit-slice modules and ripple carry; mapped board switches to inputs and LEDs to outputs for hardware testing.
- Verified all eight operations through simulation waveforms; debugged Vivado elaboration errors from mismatched module ports and confirmed corrected behavior.

12-bit ALU and Cash Register Controller | ECE 230

- Designed a Verilog cash-register controller combining a finite-state machine with a 12-bit carry-lookahead adder/subtractor built from two 6-bit stages.
- Implemented two's-complement subtraction; planned addition/subtraction test cases and compared simulated outputs with expected arithmetic results.

Autonomous RC Car | Power Distribution PCB **Nov 2025 - Present**

Electrical Design Lead & Testing Team Member

- Designed a KiCad 9 PCB integrating Raspberry Pi 5, dual L298N drivers, and VL53L1X LiDAR; led schematic capture, layout, and connector integration for four-motor differential steering.
- Separated motor/logic ground planes with a single-point net tie and bulk/decoupling capacitors; routed I2C and control lines, cleared design-rule checks, and generated Gerbers.

AI Trading Project **In Progress**

- Developing a Python state-machine strategy for breakout, retest, and second-break sequences, with mirrored call/put setups and VWAP, EMA 8, and volume conditions.
- Planning AI-based setup ranking and statistical scoring around deterministic rules, with paper/shadow trading before live execution.